

Risk of seizure recurrence after a first unprovoked seizure: a prospective study among Jordanian children

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Purpose: There is wide variation in the reported recurrence rate after a first unprovoked seizure in children. We investigated the risk of recurrence after a first unprovoked seizure in Jordanian children and the risk factors associated with increased recurrence rate.

Methods: All consecutive patients aged 3 months–14 years who presented with their first unprovoked seizures between January 1997 and 2000, were included in a prospective study and followed up for 3 years for possible recurrence. Of the patients studied, there was slight male predominance (56.6%) and 55% of them were 2–9 years of age. Generalised seizures were reported in 75% and the remaining 25% had partial seizures. The duration of seizure was 1–4 minutes in 59%. Family history of epilepsy was positive in 31% and parental consanguinity in 32%. The role of these factors in increasing the risk of recurrence was also investigated.

Results: Two hundred sixty-five patients were included in the study and continued follow up for 3 years. Ninety-eight (37%) of them experienced seizure recurrence. Among the predictor factors for recurrence, partial seizure ($P = 0.003$) and positive family history ($P = 0.000$) were associated with a statistically significant increased risk. Sex, age, duration of seizure and consanguinity were not associated with increased risk of recurrence.

Conclusion: Thirty-seven percent of the children studied experienced a second attack after a first unprovoked seizure over the 3 years follow up period. The risk of recurrence was significantly higher in children with a partial seizure (55%) and among those with a positive family history of epilepsy (59%). Age at first seizure, sex, duration of seizure and consanguinity were not significantly related to the risk of recurrence.

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INTRODUCTION

It is important to know the risk factors for recurrence following a first unprovoked seizure in children despite, nowadays most experts think for most children no indication exists to start therapy with antiepileptic drugs after a first seizure as treatment after a first seizure does not influence the prognosis of the epilepsy for those children who will have recurrence¹. There is a controversy on labelling children as epileptics after a single unprovoked seizure especially in many third world communities including Jordan since epilepsy is still regarded as a stigma^{2–4}. The reported risks of recurrence following a first unprovoked seizure in previous studies that included adults and/or children vary from 27 to 71%^{5–20}. The vast majority of these studies were carried out in the United States and

Europe. Thus, at present, the magnitude of the risk of recurrence following a first unprovoked seizure in children in many developing and third world countries is unclear. The potential predictors of recurrence that have been studied before included age at first seizure, duration of seizure, seizure type (partial vs. generalised), a family history of epilepsy, neuro-imaging and EEG^{5,6}. Consanguinity, which is widely practiced in Jordan²¹, lead to an increase in the incidence of monogenic recessive disorders. Its role as a predictor for recurrence has not been assessed before.

This prospective study was conducted to assess the risk of seizure recurrence following a first unprovoked seizure in Jordanian children and the role of number of potential risk factors including consanguinity. Our findings were compared with those from previous western studies.

METHODS AND MATERIALS

Setting

The study population consisted of a consecutive series of children 3 months–14 years of age with a first unprovoked seizure seen at Princess Rahma and King Abdullah Teaching Hospitals, the main two primary and tertiary centres in the City of Irbid, and King Hussein Medical Center, which is one of the main primary and tertiary medical center in Amman between January 1997 and January 2000. Children included in the study were allocated by a consultant neurologist after carefully excluding other non-epileptic paroxysmal events either during their hospital admission for their first unprovoked seizure or during their first visit to the Child Neurology Clinic within two weeks from their first seizure.

Children with status epilepticus and those with a cluster of seizures within 24 hours were considered to have one seizure and were included in the study. Adherent to the guidelines for epidemiologic studies on epilepsy²², and based on clinical judgment without EEG and neuro-imaging studies, children with typical absence seizures, myoclonic seizures, infantile spasm, previous unprovoked seizures, previous febrile seizures, neonatal convulsions, acute and remote symptomatic seizures, and those already receiving antiepileptic treatment were excluded from the study. The department general policy was not to initiate long-term antiepileptic drug therapy after a first unprovoked seizure and unfortunately not to perform EEG, CT scan, and/or MRI routinely unless indicated because of limited financial reasons. Routine EEG and CT/MRI if not mandatory after an uncomplicated first seizure in children but would have been important for the purpose of this study as part of a risk factors, seizure classification and for careful exclusion of children with acute or remote symptomatic seizures.

Definitions

An unprovoked seizure was defined according to the criteria outlined by Hauser *et al.*¹¹ as a seizure occurring in the absence of acute precipitating central nervous system insult, systemic metabolic dysfunction, or history of prior neurologic insult.

Status epilepticus was defined as a seizure lasting for more than 30 minutes or a cluster of seizures lasting for more than 30 minutes between which consciousness was not regained²³.

The type of seizure was classified according to the Revised International Classification of Epileptic Seizures²³.

A family history of seizures was defined as a history of an unprovoked seizure in a first degree relative [parents and siblings].

Parental consanguinity refers only to first cousins and double first cousin's marriages.

Recurrence was defined as any unprovoked seizure occurring more than 24 hours after the first seizure.

Patient evaluation and follow up

Detailed data were collected on a special data collecting sheet developed for the purpose of this study. Collected data included age, sex, seizure characteristics as generalised or partial, duration of seizures, history of prior seizures, history of prior neurological insult, history of neonatal convulsions, family history of epilepsy, and parental consanguinity. Physical and neurological examination was performed on all children. The standard practice of our paediatric neurology unit was not to perform electroencephalogram (EEG) on children with first unprovoked seizures with a few exceptions like status epilepticus.

Computed tomography or magnetic resonance imaging was also not routinely performed in our unit because of the limited financial resources except in cases with abnormal findings during the neurological examination and in partial seizures. Metabolic tests including serum sodium, calcium, magnesium, glucose, urea and creatinine were done to all participants.

After enrollment, patients were followed up every 3–6 months unless needed earlier. Ascertainment of seizure recurrence was done by history during each visit. Patients were followed-up from their enrollment into the study and for 3 years.

Predictors of recurrence

Certain factors were analysed to assess their roles as possible predictors of recurrence which included, age of the child at initial presentation, sex, family history of epilepsy, type of seizure whether partial or generalised, duration of seizure and parental consanguinity. EEG and CT/MRI abnormalities are important risk factors but unfortunately were not assessed because of financial limitation.

Date management and analysis

The data were entered and analysed using Epi-Info version 6 software. The data were checked for errors using range and logical checks. The odds ratio (OR) and its 95% confidence interval (CI) for certain presumed risk factors of seizure recurrence were

calculated. The Fisher's Exact 95% confidence limits were obtained because of the small numbers in some of the cells. Cross tabulations were performed with recurrence of seizures as the dependent variable and other study variables as the independent variables. The probability of seizure recurrence was calculated at 6, 12, 24 and 36 months following the first seizure.

Ethical considerations

An informed consent was obtained from the child's legal guardian before administering the interview. Interviews were conducted in a private setting and confidentiality of the information was assured. Identifying information was not included in the computerised data and only the researcher himself had access to such confidential information.

RESULTS

There were 265 children (150 males and 115 females) enrolled in the study and continued 3 years of follow up. Thirty-six children presented with a first unprovoked seizure but did not continue their regular follow up and therefore were excluded from the study.

Overall recurrence risk

Of the 265 children enrolled in the study, 98 (37%) experienced a recurrence within 3 years of follow up. Table 1 details the characteristics of children enrolled in the study regarding their age, sex, type and duration of seizure, family history of epilepsy and parental consanguinity. The overall recurrence was 34 (13%) at 6 months, 66 (25%) at 12 months, 87 (33%) at 24 months and 98 (37%) at 36 months.

Table 2 details the studied predictors for recurrence, partial seizures (OR = 2.8, 95% CI 1.5–5.1, $P = 0.003$) and a positive family history (OR = 3.9, 95% CI 2.2–7.0, $P = 0.000$) were the only statistically significant predictors for recurrence of seizures.

DISCUSSION

The prognosis and treatment of patients with epilepsy are best determined by the accurate classification of the epileptic syndrome, which may be difficult or impossible to make in most cases at or after the first seizure in the absence of EEG and neuro-imaging. Previous studies have shown wide variation in recurrence rates in those who have experienced a first unprovoked seizure. This is the first report to address

Table 1: Characteristics of children enrolled in the study.

Variable	Frequency	Percentage
Sex		
Male	150	56.6
Female	115	43.4
Age (years)		
<2	37	14
2–4	61	23
5–9	85	32
>9	82	31
Type of seizure		
Generalised	198	75
Partial	67	25
Duration of seizures (minutes)		
<1	32	12
1–4	156	59
5–9	48	18
10–15	19	7
>15	10	4
Family history		
Positive	83	31
Negative	182	69
Consanguinity		
Positive	86	32
Negative	179	68

the likelihood of recurrence following the first unprovoked seizure in children and assessing the potential risk factors for recurrence in Jordanian children. However, the 36 (14%) of children excluded from the study as they were lost to follow up, may have substantially altered our final result.

In this prospective study in children, the risk of recurrence after 3 years of follow up was 37%, a figure very close to Shinnar *et al.* prospective study reported in 1990 in children up to the age of 19 years when followed up for 3 years duration⁶ and less than the 42% reported by the same authors in 1996 when follow up duration extended to 6 years⁵. A second unprovoked seizure in our study was experienced in 25, 33 and 37% at 1, 2 and 3 years after the first attack, respectively, a figure much lower than the Kaplan–Meier procedure estimate of recurrence of 64% at 5 years reported by Lizana *et al.* in children⁸.

Hauser *et al.* in a hospital-based study on all age group reported that the probability of a second attack was 10, 24, and 29% at 1, 3, and 5 years, respectively, in those without CNS insult and 26, 41, and 48%, respectively, in those with a record of such insult¹⁰. Elwes *et al.* reported a much higher recurrence rate in all age group, but his sample was heterogeneous with poor follow up¹². Our rates of recurrence are slightly lower than the rates of adults^{11, 13, 19}. Exclusion of children with history of prior seizures, acute symptomatic and possible remote symptomatic seizures based mainly on clinical judgment may explain the reason of having a low rate of seizure recurrence.

Table 2: Potential risk factors for seizure recurrence.

Variable	Recurrence	No recurrence	Odds ratio	95% CI	P
Sex					
Female	40	75	1.0		
Male	58	92	1.2	0.7–2.0	0.51
Age (years)					
<2	10	27	1.0		
2–4	23	38	1.6	0.6–4.4	0.28
5–9	36	49	2.0	0.8–5.0	0.11
>9	29	53	1.5	0.6–3.8	0.37
Type of seizure					
Generalised	61	137	1.0		
Partial	37	30	2.8	1.5–5.1	0.003
Duration of seizures (minutes)					
<1	12	20	1.0		
1–4	59	97	1.0	0.4–2.4	0.97
5–9	15	33	0.8	0.3–2.1	0.56
10–15	7	12	1.0	0.3–3.7	0.96
>15	5	5	1.7	0.3–8.7	0.48
Family history					
Negative	49	133	1.0		
Positive	49	34	3.9	2.2–7.0	0.000
Consanguinity					
Negative	66	113	1.0		
Positive	32	54	1.0	0.6–1.8	0.96

Studies using different methods for identifying and recruiting patients have reported much higher recurrence risks^{7,12,14,15}. In a meta-analysis of 16 reports on the prognosis of first seizure by Berg and Shinnar⁷, the three methodological factors that most influenced the variation in the recurrence rate were the study selection criteria, retrospective or prospective ascertainment of patients, and interval between the first attack and the time at which risk was assessed. Even in studies with similar methodology, variation in prognosis can be due to the fact that the distribution of risk factors may differ leading to different recurrence rates.

A positive family history of seizures in at least one sibling or parent was noted in 31% of our children with unprovoked seizures. After 3 years of follow up, seizure recurrence rate was significantly increased to 59% ($P = 0.000$) among those with a positive family history of epilepsy. This significant increase in the recurrence rate was similar to Hauser *et al.* report¹⁰. None of the other studies that examine the role of family history in unprovoked seizures reported a statistically significant association between family history of epilepsy and recurrence risk^{13,15,18}. Shinnar *et al.* had reported that a family history of seizure was important predictor for recurrence only in the presence of an abnormal EEG⁶. Unfortunately, the role of EEG cannot be assessed in this study, as EEG was not routinely done after a first unprovoked seizure.

The proportion of children with partial seizures (25%) was lower than others¹¹. Excluding patients

with possible remote symptomatic seizures may explain this low figure, in addition to the fact that the first attack of benign epilepsy of childhood, which is the most common partial seizure of childhood, might not be observed by the patient's guardian, as they are mainly nocturnal. Partial seizures are more common in the remote symptomatic group than in children with an idiopathic first seizure. In the present study, partial seizures were associated with a higher recurrence (55%), $P = 0.003$. Previous reports have shown that in children with idiopathic first seizures, once the effect of the EEG is taken into account, there is no association between partial seizures and recurrence risk as partial seizures were more common in those with abnormal EEGs than in those with normal EEGs^{10,13}. However, on univariate analysis, four studies found partial seizures to be associated with a higher risk of recurrent seizures compared with generalised seizures^{6,15–17}. Without EEG recording we were unable to know how many children had partial seizure evolving to a secondary generalised seizure.

Consanguineous marriage is a common practice in Jordan reported in up to 50% of all marriages²¹ and known to be associated with increased incidence of many autosomal recessive disorders. Consanguinity as a possible risk factor for recurrence was assessed for the first time in this study but seems to have no role. This observation probably suggests that patients with monogenic recessive disorders do not constitute a big proportion of epilepsy patients as compared to

disorders with multifactorial aetiology. It has been observed worldwide that monogenic primary epilepsy is rare accounting for less than 1% of total cases of epilepsy.

Recurrence rate at 36 months was slightly more common in males than females but the differences failed to attain statistical significance. Previous reports reported no differences in recurrence risk between males and females^{6,10,13,15,17}. Hauser *et al.* reported earlier relapses in males than females but with similar recurrence rate at 3 years of follow up¹¹.

With the exception of the study by Hirtz *et al.*¹⁵ who reported a high recurrence among children less than 2 years of age, all previous reports found no association between age and recurrence as observed in the current study. Consistent with a number of previous studies^{6,10}, we observed no significant association between duration of seizures and recurrence. The usefulness of the duration of seizure is questionable as, outside monitoring units, this may not be reliable or reproducible.

In conclusion, 37% of the Jordanian children studied experienced a second attack after a first unprovoked seizure over 3 years of follow up. The risk of recurrence was significantly increased in children with partial seizure (55%) and those with positive family history of epilepsy (59%). Age at first seizure, sex, duration of seizure and consanguinity did not affect significantly the risk of recurrence.

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